

EART11200 - The Natural Scientist's Toolkit
Mathematics and Physics

Revision Lecture - Differentiation

- For a function, $y = f(x)$, that describes some process we are interested in, in one dimension, the derivative:

$$f'(x) = \frac{dy}{dx}$$

- It is interpreted as the rate of change of 'y' with respect to 'x'.
- Lets imagine we are observing this change over a discrete interval, such that we could write this as an approximation:

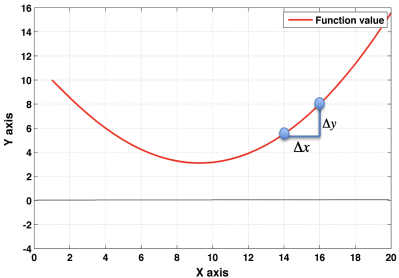
$$\frac{dy}{dx} = \lim_{\Delta x \to 0} \frac{\Delta y}{\Delta x}$$

- We looked at the quadratic function:

$$y = \sqrt{x} + \left(\frac{x-10}{3}\right)^2$$

- We want the derivative at $x = 16$.
- Assume, for now, no analytical function can be derived.
- **Euler's Backward Method**

$$\frac{dy}{dx} = \lim_{\Delta x \to 0} \frac{\Delta y}{\Delta x} \equiv \lim_{\Delta x \to 0} \frac{y_x - y_{x-\Delta x}}{\Delta x}$$

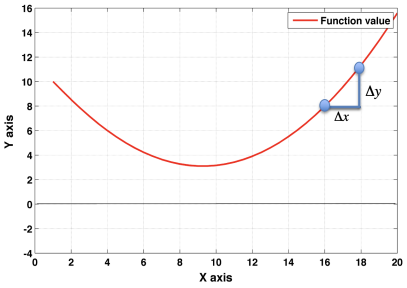


- We looked at the quadratic function:

$$y = \sqrt{x} + \left(\frac{x-10}{3}\right)^2$$

- We want the derivative at $x = 16$.
- Assume, for now, no analytical function can be derived.
- **Euler's Forward Method**

$$\frac{dy}{dx} = \lim_{\Delta x \to 0} \frac{\Delta y}{\Delta x} \equiv \lim_{\Delta x \to 0} \frac{y_{x+\Delta x} - y_x}{\Delta x}$$



- We see that using a step-size of $\Delta x = 0.25$ that the estimated gradient:
  - Euler's Backward Method: $\frac{\Delta y}{\Delta x} = -1.1071$
  - Euler's Forward Method: $\frac{\Delta y}{\Delta x} = -1.0593$
  - Central Differences Method: $\frac{\Delta y}{\Delta x} = -1.0832$
- The analytical derivative happens to be given by the following:

$$\frac{dy}{dx} = 0.5x^{-0.5} + \frac{2}{3}\left(\frac{x-10}{3}\right)$$

- At $x = 4$, $\frac{dy}{dx}(4) = -1.08333$ so we can estimate the accuracy of each method!

- For a function, $y = f(x)$, that describes some process we are interested in, in one dimension, the derivative:

$$f'(x) = \frac{dy}{dx}$$

- It is interpreted as the rate of change of 'y' with respect to 'x'. Lets imagine we are observing this change over a discrete interval, such that we could write this as an approximation:

$$\frac{dy}{dx} = \lim_{\Delta x \to 0} \frac{\Delta y}{\Delta x}$$

- We can apply similar procedures to other common functions and start populating a list of rules.
- The more you practice these procedures, the easier it will become to apply them without looking them up.
- Use your math's survival pack when needed!

Function $y = f(x)$	**Derivative $\frac{dy}{dx}$**
constant	0
x	1
x^n	nx^{n-1}
e^x	e^x
e^{kx}	ke^{kx}
$e^{f(x)}$	$\frac{df}{dx}e^{f(x)}$
$ln(x)$	$\frac{1}{x}$
$sin(x)$	$cos(x)$
$cos(x)$	$-sin(x)$

- What happens when we have a sum of functions?

$$y = 5x + e^x$$

- Well luckily, the derivative of the sum is the sum of the derivatives!
- This means we can just differentiate each term individually

$$\frac{dy}{dx} = 5 + e^x$$

Function $y = f(x)$	**Derivative $\frac{dy}{dx}$**
constant	0
x	1
x^n	nx^{n-1}
e^x	e^x
e^{kx}	ke^{kx}
$e^{f(x)}$	$\frac{df}{dx}e^{f(x)}$
$ln(x)$	$\frac{1}{x}$
$sin(x)$	$cos(x)$
$cos(x)$	$-sin(x)$

- Use the **Chain Rule** to differentiate:

$$f(x) = \sin(3x + 1)$$

- **Step 1**: Split the function and substitute

$$f(g) = \sin(g) \qquad g(x) = 3x + 1$$

- **Step 2**: Differentiate:

$$\frac{df}{dg} = \cos(g) \qquad \frac{dg}{dx} = 3$$

- **Step 3**: Apply the product rule and substitute back in:

$$f(x) = 3\cos(3x + 1)$$

The Chain Rule

$$\frac{df}{dx} = \frac{df}{dg} \times \frac{dg}{dx}$$

Function $y = f(x)$	Derivative $\frac{dy}{dx}$
constant	0
x	1
x^n	nx^{n-1}
e^x	e^x
e^{kx}	ke^{kx}
$e^{f(x)}$	$\frac{df}{dx}e^{f(x)}$
$ln(x)$	$\frac{1}{x}$
$sin(x)$	$cos(x)$
$cos(x)$	$-sin(x)$

- Use the **Product Rule** to differentiate:

$$f(x) = x\exp(3x)$$

- **Step 1**: Split the function:

$$u(x) = x \qquad v(x) = \exp(3x)$$

- **Step 2**: Differentiate:

$$\frac{du}{dx} = 1 \qquad \frac{dv}{dx} = 3\exp(3x)$$

- **Step 3**: Apply the product rule and substitute back in:

$$f(x) = 3x\exp(3x) + \exp(3x) = (3x+1)\exp(3x)$$

The Product Rule

$$\frac{df}{dx} = u\frac{dv}{dx} + v\frac{du}{dx}$$

Function $y = f(x)$	Derivative $\frac{dy}{dx}$
constant	0
x	1
x^n	nx^{n-1}
e^x	e^x
e^{kx}	ke^{kx}
$e^{f(x)}$	$\frac{df}{dx}e^{f(x)}$
$ln(x)$	$\frac{1}{x}$
$sin(x)$	$cos(x)$
$cos(x)$	$-sin(x)$

- Use the **Quotient Rule** to differentiate:

$$f(x) = \frac{3x+1}{6x-1}$$

- **Step 1**: Split the function:

$$u(x) = 3x+1 \qquad v(x) = 6x-1$$

- **Step 2**: Differentiate:

$$\frac{du}{dx} = 3 \qquad \frac{dv}{dx} = 6$$

- **Step 3**: Apply the product rule and substitute back in:

$$f(x) = \frac{3(6x-1) - 6(3x+1)}{(6x-1)^2} = -\frac{9}{(6x-1)^2}$$

The Quotient Rule

$$\frac{df}{dx} = \frac{v\frac{du}{dx} - u\frac{dv}{dx}}{v^2}$$

Function $y = f(x)$	Derivative $\frac{dy}{dx}$
constant	0
x	1
x^n	nx^{n-1}
e^x	e^x
e^{kx}	ke^{kx}
$e^{f(x)}$	$\frac{df}{dx}e^{f(x)}$
$ln(x)$	$\frac{1}{x}$
$sin(x)$	$cos(x)$
$cos(x)$	$-sin(x)$

- Finding stationary values requires a number of steps:
  1. Differentiate $f(x)$ to find the gradient function $f'(x) = \frac{dy}{dx}$
  2. Set the derivative to zero, $\frac{dy}{dx} = 0$ and solve for x
  3. If necessary, we substitute the stationary values, x, into the function, $f(x)$.
- Once you find where the derivatives are equal to 0 you can explore regions either side to determine if the gradient is positive or negative, and where the function is increasing or decreasing.
- This can tell us whether the stationary point is a maximum, minimum and/or inflection point. But, there is a more formal way to do this!

- Differential equations have an "order".
- Up to this point, we have only differentiated functions, $f(x)$ once:

$$f'(x) = \frac{df}{dx}$$

- This is known as the **first-order derivative**.
- Differentiation can be carried out multiple times.
- By differentiating again, we obtain the **second-order derivative** which is defined as:

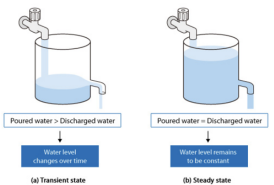
$$f''(x) = \frac{d}{dx}\left(\frac{df}{dx}\right) = \frac{d^2f}{dx^2}$$

- Second-order derivatives have interesting properties and can be used to confirm whether a stationary point is a maximum or minimum point!
- We see that for a stationary value x for the function $f(x)$:
  - If $f''(x) < 0$ the stationary value is a maximum
  - If $f''(x) > 0$ the stationary value is a minimum
  - If $f''(x) = 0$ it is not possible to conclude anything about the stationary value.
- If we're in the situation that $f''(x) = 0$, then it is potentially a saddle point. Need to investigate using the properties of the first-order derivative.

- You may not be familiar with the concept of **steady-state**.
- When a system is in steady-state, the key variables that describe the system's behavior do not change over time:

$$\frac{dV}{dt} = f(t, x_1, x_2) \implies \frac{d^2V}{dt^2} = 0$$

Comparison	Steady State	Equilibrium
Definition	Used to define consecutive reaction changes	Used to define reversible reaction changes
General equation	$A \xrightarrow{k_1} B \xrightarrow{k_2} C$ $k_1 = -k_2$	$A \underset{k_2}{\overset{k_1}{\rightleftharpoons}} B$ $k_1 = k_2$
Entropy change	Entropy increases with time	Entropy remains constant
Type of system	Open system	Closed system
Heat transfer	Transfer of heat occurs	No net transfer of heat occurs
Type of process	Kinetically controlled processes	Thermodynamically controlled processes



- In many cases, it will not be possible to calculate when a function has a value of 0. Important for stationary values.
- For example, how do you solve:

$$2x^2 + 5 - e^x = 0$$

- Use Newton's Method for Root Finding:

$$x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)}$$

- We pick a starting guess x_0 and apply this formula iteratively until we reach some threshold.

- A **scalar field** is a function of two or more variables

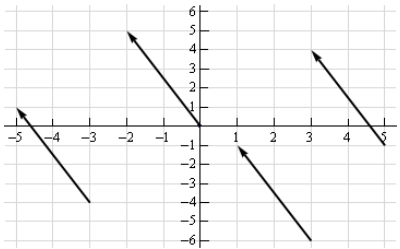
$$f(x, y) = x^2 + y^2$$

- Scalar fields represent a single value at different positions.
- They have **magnitude but no direction** and can be represented with a colour plot.
- A **vector field** is a mathematical function that assigns a vector to each point in space.
- They have **magnitude and direction** as we can determine a direction, or flow and the magnitude from the length of the arrow.

- Each line in this figure represents the same vector, but a different starting position.
- Wherever the starting position is, each vector moves 2 units to the left and 5 units up.
- If we write this procedure in vector form we write:

$$\vec{v} = \langle -2, 5 \rangle$$

- Notice the distinction in style between the variable and the brackets.
- Specifically, $\langle -2, 5 \rangle$ is fundamentally different from writing $(-2, 5)$.

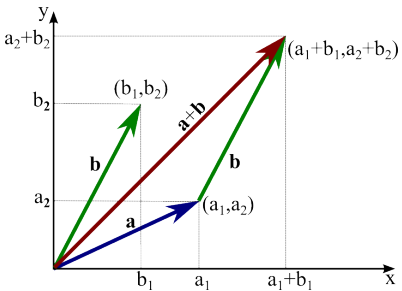


- The magnitude of a vector is calculated by taking the square root of all of the square its components (i.e. Pythagoras' theorem):

$$|\bar{a}| = \sqrt{a_1^2 + a_2^2}$$

- Vectors can be added or subtracted (add or subtract by component) or multiplied by a scalar (constant).
- Given the vectors $\vec{a} = \langle a_1, a_2 \rangle$ and $\vec{b} = \langle b_1, b_2 \rangle$:

$$\begin{aligned} \vec{a} + \vec{b} &= \langle a_1 + b_1, a_2 + b_2 \rangle \\ \vec{a} - \vec{b} &= \langle a_1 - b_1, a_2 - b_2 \rangle \\ c\vec{a} &= \langle ca_1, ca_2 \rangle \end{aligned}$$



- We now think about differentiation of functions of two (or more) variables (i.e. scalar field), $z = f(x, y)$, which now depend on x and y.
- For a one dimensional curve the gradient $\frac{dy}{dx}$ is fairly straight forward.
- When using more than 1D, we have to use something else to denote the gradient.
- The standard that is used is called 'grad', which is defined for a scalar field and can be written ∇f.
- Note that ∇f (where $f = f(x, y, z)$ is a scalar field) is actually a vector.

$$\nabla f \equiv \frac{\partial f}{\partial x}\hat{i} + \frac{\partial f}{\partial y}\hat{j} \equiv \left\langle \frac{\partial f}{\partial x}, \frac{\partial f}{\partial y} \right\rangle$$

- The gradient of a scalar field has magnitude and direction!
- Notice the change in the derivative symbol:

$$\frac{\partial f}{\partial x}$$

- Up to this point in our course, we have been using the expression:

$$\frac{df}{dx}$$

- The change is important as this is called a **partial derivative**.
- When you have a function expressed using more than one dimension, this means that when calculating a derivative in say x, you assume other variables such as y or z remain constant.

- The gradient has magnitude and direction!
- Notice the change in the derivative symbol:

$$\frac{\partial f}{\partial x}$$

- Up to this point in our course, we have been using the expression:

$$\frac{df}{dx}$$

- The change is important as this is called a **partial derivative** and we move into using **partial differential equations**.
- When you have a function expressed using more than one dimension, this means that when calculating a derivative in say x, you assume other variables such as y or z remain constant.

- Select TRUE or FALSE to the following statements about differentiation.
  (a) If $f(x) = \cos(2x)$ then $\frac{df}{dx} = 2\sin(2x)$?
- False!

$$\frac{df}{dx} = -2\sin(2x)$$

- Select TRUE or FALSE to the following statements about differentiation.
  (b) If $f(x) = e^{-2x}$ then $\frac{df}{dx} = -e^{-2x}/2$?
- False!

$$\frac{df}{dx} = -2e^{-2x}$$

- Select TRUE or FALSE to the following statements about differentiation.
  (c) If $f(x) = \ln(2x)$ then $\frac{df}{dx} = 1/2x$?
- False!

$$f(x) = \ln(2x) = \ln(2) + \ln(x) \implies \frac{df}{dx} = \frac{1}{x}$$

2023-24 Past Paper

- Select TRUE or FALSE to the following statements about differentiation.
  (d) If $f(x) = 5x^n$ then $\frac{df}{dx} = 5nx^n$?
- True!

- Select TRUE or FALSE to the following statements about differentiation.

  (e) If $f(x) = \sin(x)$ then $\frac{d^2f}{dx^2} = -\sin(x)$?

- True!

- Using Newtons method, find the root of the following function:

$$f(x) = e^x - x^2$$

correct to 3 decimal places. You may use a starting guess of $x_0 = -0.6$. Recall that we can write Newtons formula for root finding, generally, as:

$$x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)}$$

where $f'(x)$ is the derivative of $f(x)$.

- First, we need to find the derivative of $f(x)$:

$$\frac{df}{dx} = f'(x) = e^x - 2x$$

- Calculate the first guess (x_1):

$$\begin{aligned} x_1 &= x_0 - \frac{f(x_0)}{f'(x_0)} \\ &= -0.6 - \frac{0.188811}{1.748812} = -0.7079657 \end{aligned}$$

- Calculate the second guess (x_2):

$$\begin{aligned} x_2 &= x_1 - \frac{f(x_1)}{f'(x_1)} \\ &= -0.7079657 - \frac{-0.008570}{1.908577} = -0.7034754 \end{aligned}$$

- Calculate the third guess (x_3):

$$\begin{aligned} x_3 &= x_2 - \frac{f(x_2)}{f'(x_2)} \\ &= -0.7034754 - \frac{-1.5188 \times 10^{-5}}{1.901813} = -0.70347674 \end{aligned}$$

- So the root of $f(x)$ correct to 3 decimal places is $x = -0.703$.

2023-24 Past Paper

- A gas leak is emitting Butane at a variable rate. The concentration equation is given below, where G is given in micrograms per cubic metre and the x-axis given in minutes.
- The associated equation for G is also given as:

$$G(t) = 3 + 14t - 5t^2$$

- Find the time at which the maximum concentration of emitted gas occurs.

- We need to derive $\frac{dG}{dt}$:

$$G(t) = 3 + 14t - 5t^2 \implies \frac{dG}{dt} = 14 - 10t$$

- To get the maximum we set the derivative to zero and solve for t

$$\frac{dG}{dt} = 14 - 10t = 0 \implies 14 - 10t = 0 \implies t = 1.4$$

- Differentiate the following expressions with respect to x. Simplify your answer as much as possible:

$$f(x) = \cos(x)e^{-5x}$$

- **Step 1**: Split the function:

$$u(x) = \cos(x) \qquad v(x) = e^{-5x}$$

- **Step 2**: Differentiate:

$$\frac{du}{dx} = -\sin(x) \qquad \frac{dv}{dx} = -5e^{-5x}$$

- **Step 3**: Apply the product rule and substitute back in:

$$\begin{aligned} f(x) &= -5\cos(x)e^{-5x} - \sin(x)e^{-5x} \\ &= -e^{-5x}(\cos(x)e^{-5x} + \sin(x)) \end{aligned}$$

- Differentiate the following expressions with respect to x. Simplify your answer as much as possible:

$$g = \frac{x}{\sqrt{x^2 + 4}}$$

- **Step 1**: Split the function:

$$u(x) = x \qquad v(x) = \sqrt{x^2+4} = (x^2+4)^{-1/2}$$

- **Step 2**: Differentiate:

$$\frac{du}{dx} = 1 \qquad \frac{dv}{dx} = x(x^2+4)^{-1/2}$$

- **Step 3**: Apply the quotient rule and substitute back in:

$$\begin{aligned} f(x) &= \frac{\sqrt{x^2+4} - x^2(x^2+4)^{-1/2}}{x^2+4} = \frac{\sqrt{x^2+4} - x^2(x^2+4)^{-1/2}}{x^2+4} \times \frac{\sqrt{x^2+4}}{\sqrt{x^2+4}} \\ &= \frac{x^2+4-x^2}{(x^2+4)^{-3/2}} = \frac{4}{(x^2+4)^{-3/2}} \end{aligned}$$

- Differentiate the following expressions with respect to x. Simplify your answer as much as possible:

$$y = \frac{e^{-x^2}}{x}$$

- **Step 1**: Split the function:

$$u(x) = e^{-x^2} \qquad v(x) = x$$

- **Step 2**: Differentiate:

$$\frac{du}{dx} = -2xe^{-x^2} \qquad \frac{dv}{dx} = 1$$

- **Step 3**: Apply the quotient rule and substitute back in:

$$\begin{aligned} f(x) &= \frac{-2x^2e^{-x^2} - e^{-x^2}}{x^2} \\ &= \frac{-e^{-x^2}(2x^2+1)}{x^2} \end{aligned}$$

- A fuel tank has a slow releasing leak. We want to study the volume of fuel in the tank over time t which can be described using the following equation:

$$V(t) = \frac{V_0}{10}(t^2 + 10)\exp\left(-\frac{t}{2}\right)$$

where $V_0 = 10$ is the volume of fuel (in m^2) and $0 < t < 24$ (in hours). What is the rate of the leak at 2 hours? Determine the rate analytically but also compare to a rate calculated numerically using central differences method with a time step Δt of 30 minutes. Estimate the percentage error of the numerical result.

- **Step 1**: Split the function:

$$u(x) = t^2 + 10 \qquad v(x) = \exp\left(-\frac{t}{2}\right)$$

- **Step 2**: Differentiate:

$$\frac{du}{dx} = 2t \qquad \frac{dv}{dx} = -\frac{1}{2}\exp\left(-\frac{t}{2}\right)$$

- **Step 3**: Apply the product rule and substitute back in:

$$\begin{aligned}
\frac{dV}{dt} &= \frac{V_0}{10}\left(-\frac{1}{2}(t^2+10)\exp\left(-\frac{t}{2}\right) + 2t\exp\left(-\frac{t}{2}\right)\right) \\
&= \frac{V_0}{10}\exp\left(-\frac{t}{2}\right)\left(-\frac{1}{2}(t^2+10) + 2t\right) \\
&= -\frac{V_0}{20}\exp\left(-\frac{t}{2}\right)\left(t^2 - 4t + 10\right)
\end{aligned}$$

- To find the rate at two hours we evaluate the derivative at $t = 2$:

$$\begin{aligned}
\frac{dV}{dt}(2) &= -\frac{10}{20}\exp\left(-\frac{2}{2}\right)\left(2^2 - 4 \times 2 + 10\right) \\
&= -\frac{1}{2}\exp(1)\left(4 - 8 + 10\right) \\
&= -\frac{6}{2}\exp(1) \\
&= -1.10363
\end{aligned}$$

- We can also find the rate at 2 hours numerically using the central differences method. We use a timestep of 30 minutes, so evaluate the Volume V_0 at $t = 1.5$ and $t = 2.5$

$$\begin{aligned} V(1.5) &= \frac{10}{10}(1.5^2 + 10)\exp\left(-\frac{1.5}{2}\right) = 12.25\exp(-0.75) = 5.78649 \\ V(2.5) &= \frac{10}{10}(2.5^2 + 10)\exp\left(-\frac{2.5}{2}\right) = 16.25\exp(-1.25) = 4.65570 \end{aligned}$$

- Calculate the rate numerically:

$$\frac{\Delta V}{\Delta t}(2) = \frac{4.65570 - 5.78649}{2.5 - 1.5} = -1.13079$$

- The rate has an error of 0.02716 which as a percentage is:

$$100 \times \frac{-1.13079 + 1.10363}{-1.10363} = 2.46\%$$